

Team USA in Switzerland

Physics fights, foosball rooms, Swiss cows, ice cream, and five brilliant teenagers

Prologue: Before Switzerland Was Switzerland

Long before Team USA ever found itself wandering around Switzerland looking for affordable food, rooms with working outlets, and Swiss people willing to smile at us, there was the small matter of actually fielding a team.

This whole thing did not begin at the airport. It began months earlier with students submitting rough drafts, coaches and board members reading papers, students revising problems, and me slowly realizing that international physics travel is not a trip. It is a five-month group project where the group is the entire country and somehow I am still the one buying the granola bars.

The selection process ran from early February into late March. Students submitted work. We reviewed it. We gave feedback. We selected finalists. We had interviews. We sent invitations. We sent the much less enjoyable emails to students who did not make the team. Then, almost immediately, the job changed. The physics was still there, but now it was mixed with fundraising, family contributions, plane tickets, hotel reservations, passports, food budgets, tournament registration, and the special joy of fronting money while hoping the money eventually came back from somewhere other than my personal checking account.

By late June, we had a team, we had problems, we had presentations, we had an international scrimmage with South Africa and Great Britain, and we had just enough confidence to believe that this group could do something special. The five students were Irene, Ivey, Bridget, Val, and Kurt. Noreen and I were the adults, at least by birth certificate. The students were the physics team. We were the support staff, logistics department, food committee, budget office, emotional regulation unit, and, on more than one occasion, the people trying to figure out if the tram tickets we had just bought were actually useful for the tram we were currently standing next to.

This is the story of that trip. It is about physics, but it is mostly about the kids. It is about watching students become a team in real time. It is about brilliant teenagers doing hard things in public. It is about good decisions, questionable decisions, and one or two decisions that made everyone want to stab me in the neck with a Sharpie. It is about being proud of students in a way that is difficult to explain unless you were there.

Also, there is a Swiss cow.

Chapter 1: Day Two - Physics, Foosball, and Our First Swiss Boundary

Day Two began with a plan so responsible and adult-like that I barely recognized it as one of mine.

Noreen and I were going to get up early, go to the grocery store right when it opened at 8:00, and buy breakfast for the kids. We had polled them ahead of time, because apparently teenagers will tell you what they want to eat if you ask them directly instead of just throwing granola bars at them like zoo animals. The requests were reasonable: Greek yogurt, a little cereal, pastries, and fruit.

So we bought all of that. It was relatively inexpensive, and we were able to get through breakfast at about \$10 per kid. For one brief shining moment, I felt like we were running a highly efficient international physics operation instead of a traveling circus with graph paper.

After breakfast, we had a logistics meeting. We talked about the week, the schedule, the workload, roles and responsibilities, and how we were going to survive the tournament without losing a laptop, a passport, or the will to live. Then we started right away with a practice fight.

Val presented first, and she presented one of our strongest problems. This problem was complete in the way you dream a problem will be complete before an international tournament. The theory was solid. The experimental design was excellent. She had collected a ton of data. The data reinforced the theory. It was the kind of problem that makes you feel like maybe, just maybe, the whole operation is not being held together by binder clips and prayer.

Kurt opposed her. Kurt, as I had already started to understand, brought a different kind of experience into the room. He had worked with the German team. He had seen oppositions. He had a template. He had a system. He knew how to ask questions in a way that moved the fight forward without turning the whole thing into a polite physics mugging. He and Val had an excellent exchange.

Bridget and Irene handled the review, and Bridget did the lion's share of the speaking. She was clear, thoughtful, organized, and really good. If I were attempting to judge the whole fight impartially, which I cannot because these are my kids and I am wildly compromised, I would have scored it very well. In fact, if you told me that we could get that same quality of report, opposition, and review in every fight all week, I would sign for it immediately. We would be in medal position.

Everything about the fight was great. Except for the facilities.

We had found a room in an adjacent hotel, right off the lobby. It had a foosball table, a pool table, a dartboard, and a giant screen television for watching World Cup games. Basically, it was a game room that nobody was using at 10:00 in the morning. Naturally, we took it over completely.

We spread out over every table in the room. We used the portable projector Rob Zetley had given us and projected physics slides onto the wall. We conducted a full practice physics fight while navigating around a foosball table and competing with background music that sounded like bad cover versions of 1980s songs you might hear in an elevator if the elevator itself had given up.

Every time we unplugged the sound system so the students could hear each other, the facilities guy appeared and plugged it back in. He had the timing and persistence of a Swiss ninja whose only mission was to defend mediocre pop music.

Noreen, meanwhile, had decided to take on the role of the innocent American who violates all the rules and then pretends we did not know there were rules. She wandered into a little kitchenette area to use the sink and grab plates, and eventually the same facilities guy caught her and chastised her for being somewhere she was not supposed to be.

Kurt, who speaks German, later reported that as the man walked away he said something along the lines of these ugly Americans touching things they did not have permission to touch.

Noreen took this as a point of pride. We had found a boundary.

Now the plan is to win this man over completely. We are going to buy him a gift. We are going to get a picture with him. By the end of the weekend, there may be a hug. He does not know this yet, but he is part of the team now.

After the first fight, we triaged it. We talked about what worked, what did not work, what needed tightening, and how to make the whole process cleaner. I had intended to do a second fight in the morning, but by the time we finished the fight and the feedback, it was nearly 12:30. So we broke for lunch.

Lunch was at Coop, a buffet or make-your-own lunch place across the street that compares loosely to Wegmans, if Wegmans were smaller, more Swiss, and less emotionally invested in macaroni and cheese. It was not great food, but it was in our wheelhouse and in our budget.

In the afternoon, we ran our second fight. Irene presented Oil Flow. Ivey opposed her. Kurt and Bridget worked on the review. Irene had not exactly volunteered. In fact, when I asked her if she wanted to present, she said she did not really want to. Then I asked Ivey, and she said she did not really want to either. I explained that while I respected everyone's feelings, someone did, in fact, have to present the problem.

So the students requested a random drawing. We drew lots. Irene won, which in this context meant she lost. And then she handled it beautifully.

That second fight also went well. Kurt continued coaching the team on opposition strategy, and the students kept learning from him. His German-team experience was incredibly useful. He had a way of making the process more structured and efficient, which is exactly what we needed.

By the time we finished the second fight, it was almost 4:30. My original plan had been to complete five fights, one for each student. We completed two. So, yes, my plan was adorable.

But the two fights we completed were excellent. The kids were learning fast. They were figuring out their roles. They were starting to trust each other. It had the emotional energy of a middle school dance: everyone wanted to participate, everyone wanted feedback, everyone wanted to be involved, but nobody wanted to be the first one on the dance floor. So I was basically standing in Switzerland forcing brilliant teenagers to slow dance with physics.

Our interactions with the Swiss people continued to be a long-term project. We smiled. We said hello. We tried to be friendly. In general, the Swiss people seemed to want us to leave them alone. The only people openly friendly to us so far were the three-year-old twins who wanted to practice high-fiving and the grocery store employees who are paid to be nice to us. But we were not giving up.

For dinner, we walked to a Thai restaurant, because public transportation in Switzerland is not cheap, and our budget was a real thing. If we could get through a day of feeding the kids for about \$500, we could stay on track. Yesterday was a little heavy because of transit. Today was a little light, which meant the most important thing of all: maybe we could afford ice cream all week.

Dinner also produced one of the most revealing conversations of the trip. Kurt described his experience with the German physics team. According to Kurt, the German team essentially has an unlimited food budget. When they go out for dinner, they go to very nice restaurants and are told to keep it under about \$100 per kid. He said it was some of the best eating of his life.

The German team also brings a support team of eight to ten former IYPT competitors. Every student may have one or two experienced physicists helping them. They get around 1,000 applicants, pick the best 12, run five weekend-long mini-camps, eliminate four students, take five to IYPT, and send the next three to a related Austrian tournament. If Germany makes the finals, the students can receive academic scholarship money that pays them around \$1,000 a month in college.

Germany shows up with a machine. We show up with a portable projector, a food spreadsheet, some Swiss francs, and the audacity to believe we belong in the room.

Later that same night, Ivey was afraid to ask me to buy her a toothbrush because she thought we could not afford it. To be clear, we could afford the toothbrush. But that pretty much captures the difference

between what Germany brings to the tournament and what we bring. Germany arrives with infrastructure. We arrive with determination and a very strong commitment to reasonably priced yogurt.

After dinner, we walked through Old Town. The architecture was beautiful. The kids took a ton of pictures. We spent part of the evening giving Val ridiculous prediction challenges. I asked her to predict the first high-five in recorded history. She somehow landed on October 2, 1977, when Dusty Baker hit a home run for the Los Angeles Dodgers and high-fived a teammate. I was alive at the time and did not know this. I have no idea how Val did.

Then we moved on to the first fist bump, and Val again got the year and the scenario. Then I asked when the first Hawaiian shirt was worn and when it was first considered a Hawaiian shirt. Bridget and Val went through a complicated derivation involving war, travel, veterans, and Asia, and Bridget landed within a year or two.

These kids are ridiculous. I mean that as the highest possible compliment.

Finally, we made the mandatory ice cream stop. Bridget informed me that the real pressure this week was not the international physics tournament. The pressure was that when I take another picture of the kids eating ice cream, it needs to be better than last year's ice cream picture. Apparently, I have repeatedly chosen terrible pictures of her and damaged the carefully curated Bridget Whitfield Instagram brand.

So now, in addition to coaching physics, managing budgets, buying groceries, finding rooms, and creating mild international incidents, I am also responsible for elite-level ice cream photography.

Chapter 2: The Kids Decide to Oppose Me

The plan for the next day was two more presentations in the morning. Then the kids decided they needed practice working together as an entire team against one common opponent. Naturally, that common opponent had to be me.

I was going to take one of the problems and present it as the reporter. The five of them were going to work together and oppose me. I was pretty sure they were going to kick my ass. Individually, all five of them are smarter than I am. Together, they are basically Physics Voltron.

My first hurdle was going to be talking fast enough and briefly enough to get through the entire presentation. Anyone who has ever received one of these updates knows that brevity is not exactly my defining gift.

At some point I also posed the most important question of the day: of the five students - Irene, sophomore golf expert from Los Angeles; Ivey, rising junior academic superstar from Princeton, New Jersey; Bridget 'Boom Boom' Whitfield, former All-State musician and future Haverford alum; Val 'Valedictorian' Wakeman; and Kurt, former German superstar now fully Americanized - which one was most interested in Egypt's likelihood of winning in the World Cup that evening? For that matter, which one was most interested in the World Cup altogether?

I will not pretend that this was academically essential. But when you are coaching international physics in a Swiss hotel game room, you have to find joy where you can.

Chapter 3: Moving Day

In IYPT terms, moving day is the day when things either start to open up or start to fall apart. It is two fights in one day: a morning fight, about an hour for lunch, and then another fight immediately afterward. They do this so the following day can be used for the cultural excursion. In our case, that meant hiking in the Alps, which sounded lovely in the abstract and increasingly questionable the more I thought about my knees.

But before the kids could touch grass, hang out, take pictures, and maybe enjoy Switzerland for the first real day, we had to get through a drama-filled day of physics shenanigans.

After Round Two, we had another solid day but dropped slightly into a tie for 11th place. In the morning, we were in a four-team fight against Germany, Switzerland, and Serbia. A four-team fight takes an extra hour, which meant our already intense two-fight day was going to be longer than normal. Germany and Switzerland were both top-five teams. Serbia was right behind us in the standings. So the schedule was, to use the technical term, not gentle.

For this fight, Germany was opposing us. Germany was the second-ranked team in the competition going into the round, and because Kurt is friends with a lot of the German kids, we caught wind that they intended to oppose Bridget's Singing Ruler problem. Bridget went to bed knowing she was going to be tested. She woke up with a steely determination.

When Bridget showed up to the fight, she crushed it. She batted away a lot of their attacks. Some landed, because Germany was Germany and there is a reason they were sitting near the top of the tournament. But Bridget held her own and earned a solid 5.2. That is a real score against a real opponent in a real fight, and she earned it.

In another stage, we had Irene review a fight between Switzerland and Serbia. Irene had been quiet early in the tournament, and I had told her before the competition began that I was intentionally going to keep her on the sidelines for the first couple of rounds so she could see what this world looked like before we threw her into it. By the third or fourth round, I wanted to slip her into action.

And we did. The funny thing was that Irene did not know the problem she was reviewing. But she had a solid team behind her, and they worked through it with her. The only real criticism I heard was that she was quiet and needed to bellow a little more. Other than that, she absolutely rocked it and earned scores around 5.4, which we will take every day of the week and twice on moving day.

Then Val opposed Serbia on Autumn Coin. This was a problem we had attempted to solve. We had gotten it maybe 70 or 75 percent of the way done and decided it was not close enough to present. So we had put it on the shelf. But Val understood enough of the problem to carry on an intelligent conversation with the reporter. And boy, did she ever. Despite not having completed the problem herself, she earned three 8s, a 7, and a 6 for a 7.5 average. Our total score for the round was exemplary, and when it was all said and done, Team USA climbed back into the top 10.

Things looked rosy. Which, in retrospect, should have made me suspicious.

Chapter 4: The Sharpie Moment

After the morning fight, the kids would normally hang out, trade pins, share war stories, connect on Instagram or TikTok or whatever teenagers are doing now, and generally enjoy the international student part of the tournament. Instead, I had to kick them out, get them to lunch, and get them back for the afternoon fight.

The afternoon fight was just as hard. We were going against another top-10 team and Thailand. At this point, I need to get into the weeds of the competition, because the kids thought I made a mistake. Perhaps I did. I talked it out with Noreen after her swim in Lake Zurich, and I still think what I did was defensible. But it was scary in the moment, and for about an hour I am pretty sure the kids wanted to stab me in the neck with a Sharpie.

The backstory goes to 2019 in Warsaw, Poland. That year, I thought we had a stacked team. We arrived with nine completed problems and two works in progress. I was terrified of taking a penalty, so I allowed students to present problems even when I knew the work was not complete. One of those reports received a score so low I had never seen anything like it in competition. We finished just outside bronze medal position, and I have been questioning that decision ever since. Should I have rejected the problem, taken the risk, and tried for something better?

Well, that exact kind of decision came back in Switzerland.

At that point in the competition, we had five problems remaining. Two were complete and likely to score well. Three were the kind of 75 to 80 percent complete reports that scare me because they look good until you see them next to the top international teams. After watching other countries present related work and seeing the scores they earned, I felt confident that if we presented those incomplete reports, our chance at silver would disappear.

So I did a little game theory. I looked at what our opponent could call. I looked at the rejection rules. I calculated that we could reject several times and still stay alive. The probability of being kicked out of the tournament felt close to zero. However, close to zero and zero are not the same number.

I told Val we were going to reject everything except the one complete problem we wanted. The other team called a problem. We rejected it. They called another. We rejected. Then another. We rejected. Each rejection was followed by a minute of their team huddling, deciding what to call next, and then us rejecting again. Eventually, we were at the point where if we rejected one more time, we would be eliminated from the tournament and all of our work would be for nothing.

We expected them to call Electrical Dampening, one of our most complete reports, done by Kurt, one of our strongest students. We wanted Kurt on the stage. But they avoided it. I do not think it was malicious. I think they were simply trying to avoid a problem they did not want to see. Unfortunately, their strategy and our strategy were now wrestling in the middle of the room while my blood pressure tried to leave my body.

Then they called Self-Starting Siphon.

This was not what we wanted, but it was not immediately a tragedy. Irene had worked on Self-Starting Siphon and had done nice work. But she had been working largely alone in Korea, and the report was less complete than the problems I wanted to present. In fact, a week earlier I had told Irene to stop working on it because I did not think we would ever present it.

Now we had a choice: Irene had to wing it, or we were out of the tournament.

Irene is tough. Irene is smart. I am sure she had moments of doubt, but you never saw them on her face. There were technical problems. She worked through them. She stood up and gave a solid 12-minute presentation on a problem she had not practiced in more than a week. She was a little unfamiliar with the slides. The physics was light in places. But we had a presentation, and all we needed was for her to get threes so we would still have a fighting chance to win a medal.

Wouldn't you know it, that is exactly what Irene did. She was awesome.

I think she found something inside herself that she did not know she had. When the opponent pointed out flaws, she did not simply accept them as fact. She fought back. Often the opponent was right. The flaw

was the flaw. But in that moment, in that arena, knowing she was going to take a beating, Irene stood there and basically said, please sir, may I have another.

I put that child in a terrible situation. She stood up and said, no problem.

After the fight, everyone was a little salty. Val probably still thought I made the wrong decision. Kurt thought I made the right decision. I was not exactly sure what Bridget thought. Irene may have had physics PTSD for the next several hours. Ivey was just thankful it was not her.

We went to dinner and waited for scores. I tried to give the kids a pep talk. We talked about what had gone well, what they had done, and how the team had rallied around Irene when I put her in that awful situation. These kids were a team. They were picking each other up. And that day Irene picked me up, which is no small thing for a 5-foot-1 16-year-old girl from Los Angeles.

When the scores came out, we had fallen. We took a penalty for the rejections and a penalty for the threes. We finished with 27 points, about 10 points lower than expected, and dropped to 16th place. That was not a tragedy. We were still in bronze-medal position. We still had a lead over the teams that could knock us out. We still had a fight the next day where we were positioned to do well.

But I spent a lot of time beating myself up. If I had played it conservatively, we almost certainly would have protected bronze. But the kids had been in silver position. They had a chance to achieve the goal they had set at the beginning of the week. I was not willing to give up on that without trying.

After dinner, the kids went for a swim. The questions for the next fight were announced. And immediately, they started working. They were locked in. All week we had talked about being a team and about the idea that it was not about who was on the stage, but about what the team did together. I may or may not have made a mistake, depending on who you ask. But I was counting on this team to pick me up.

Chapter 5: The Forgot-to-Mention Department

In the continuing tradition of me forgetting item after item after item, there were two important pieces from the day of the multiple rejections that I managed to leave out at first. This is what happens when you try to summarize international physics, student emotions, tournament strategy, meal logistics, and personal guilt in one travel update. Things get missed.

The first was Ivey's review. The fight actually began with a review, and Ivey had not yet been on stage that week. We had prepped her for Sweet Monochromator, a problem involving a column of sugar solution and polarizers on each end used to separate wavelengths of light. The physics is technical and specific, but our apparatus was cool, and Val had already done very well reporting on it earlier in the week. We decided Ivey should review because she had done such a good job during boot camp and we wanted to give her a chance to thrive.

And thrive she did.

Last summer in Sweden, Ivey was a little shy. She was reluctant to speak up. She was still relatively new to the United States and dealing with a language barrier. None of that was visible in Switzerland. In her review, she killed it. Her questions were on point. She was loud. She was forceful. She made eye contact with the opponent, the reporter, and the judges. Then she sat down and crafted a terrific review of the report, the opposition, and the structure of the whole exchange. It was a fantastic job, and she earned scores of 5.6. That gave us confidence going into the next round.

The second thing I had forgotten was Kurt's opposition on Flip Over Flip. This was a problem he had completed and would present the following day, and I thought it was one of our best remaining reports.

While listening to the other team's report, Kurt successfully digested the mathematical model, identified where it matched his own thinking, and pointed out where he believed the reporter had made incorrect assumptions. He had perhaps the best conversation he had had with a reporter all week.

Earlier, judges had given him feedback that he was a little too curt - which is funny because his name is Kurt, and apparently even his criticism came with a built-in pun. This time he did a much better job interacting with the people in the fight, and he delivered solid scores in the sevens and eights. It was an outstanding opposition.

The other thing worth saying is that our schedule had been brutal. Of the 13 teams we had competed against, we had been outscored by 11 of them. On the surface, that sounds like bad news. But in reality, it showed how hard our draw had been. We had played against most of the top 10 and many of the next five, and we were still sitting in bronze-medal position. In the 10 years I have been doing IYPT, this was the hardest schedule we had ever faced. I was incredibly proud of the kids for putting us in that position.

Chapter 6: Day Eight - Touching Grass, Climbing Mountains, and Kissing Cows

Day Eight had no official physics. It was cultural excursion day, and the organizers planned for us to go hiking in the Alps. This was, quite frankly, the best cultural excursion I have experienced at IYPT. It was simple. It was not organized fun, which is usually the worst kind of fun. They just brought the kids to a beautiful place in Switzerland and gave them time to take selfies, hang out, and be teenagers in the mountains.

The day started with a chartered bus that took us to a resort called Stross, or Stoos, or something close enough that I am not going to make Swiss spelling the hill I die on. When we arrived, they gave us a bag lunch and a choice between vegetarian and vegan. Everyone on our team chose vegetarian, which turned out to be a cheese sandwich on a pretzel roll, a powdered-sugar-covered strudel, and a bottle of apple juice. On paper, that does not sound like the greatest lunch ever. In practice, the sandwich was pretty darn good, everyone ate the strudel, and not everyone was equally impressed with the apple juice.

In the winter, the mountain is a ski resort. In the summer, farmers take cows up there and let them graze, and the hiking trails wind through the same spaces. We took a ski lift to the middle of the mountain and then had to choose between the strenuous hiking path and the easy path. Noreen had decided she wanted the hard path. I expected most of the kids would want the hard path too, but I had quietly decided that if any kids wanted the easy path, I would selflessly volunteer to walk with them. I am old. I did not need to prove anything to a Swiss mountain.

Naturally, every kid wanted the challenging path. So Noreen and I muscled up and followed them.

This ski area required a first tram up a very steep incline to the middle of the mountain. Then we took a second tram to the top. From there, the hike followed the ridge from one tram station to the other. It was about 5 kilometers if you could walk in a straight line, which of course you could not. We went up and down several times, with repeated climbs of several hundred meters. By the time Noreen checked her steps, we had something like 13,000 steps on the afternoon, and many of them felt personally targeted.

By the end of the walk, we were all beaten. We took the tram down to the valley, got on the bus, and headed back into town. But the highlights were worth every step.

First, Irene did not feel well. She did not eat, and we kept pumping her full of water and whatever over-the-counter pain reliever we could find. I was quietly hoping she might take the easy path and get me

off the hook. Instead, Irene refused to do that. In no time, with no food in her stomach and medicine in her system, she was walking across the Alps like a champ.

Second, Noreen and I realized how much we trusted this group and how different their priorities were. Val was project motivated and wanted to do the hike. Some of the other kids were more interested in getting pictures with the Alps in the background, and to be fair, some of those selfies were fantastic. At various points, the team was spread across the Alps: Irene and Val were pushing toward the finish line, Bridget and Kurt were taking in the sights, and Noreen and I were trying to lead from the middle with Ivey.

Ivey is always spelled I-v-e-y, and I am including that here because apparently I need to say it at least once per document.

Noreen and I were very careful with our footing. We had visions of twisting an ankle, tumbling down the mountain, being airlifted out, and appearing on the local news as dumb Americans doing dumb American things. We were committed to three points of contact. Ivey, meanwhile, walked the same trail while eating a sandwich with one hand, taking pictures and selfies, texting, and making phone calls with the other. She never stumbled. She never looked concerned. She treated the Swiss Alps like a hallway between classes. I do not know about Noreen, but I decided this meant I am very, very old.

The other fun thing about traveling with me - and my own children will confirm this - is that I like to make kids do stupid things. When my oldest daughter was around 12, I used to pay her a dollar to eat strange things. This worked well until she learned that the tasty thing she had just eaten was fried calamari and then threw up all over the restaurant. My younger daughter, having seen the damage, became much better at ignoring me when I said things like, 'Just try a bite. I'll give you a dollar.'

Ivey, unfortunately for her, was not fully aware of this character flaw. Remember the cows grazing on the mountain? I had a Swiss franc in my pocket, and I told Ivey I would give it to her if she rode a cow. She wisely refused, even though the cow was lying down, which was probably for the best because she is smarter than I am. But she immediately countered that she would kiss the cow.

So somewhere there exists a photo of a 16-year-old American physics team member, visiting from Beijing, kissing a Swiss cow on top of the Alps. I really love this girl.

Back in town, we ate in the cafeteria. The meal was meatloaf, and it was pretty tasty. Afterward, the team decided there needed to be more ice cream, because apparently we had developed a policy. We let Val take total control because that is what Val does. She led us through the mean streets of Zurich, up cobblestone streets and alleyways, until we eventually landed at a Swiss gelateria. It was some of the best ice cream I have ever had.

Finally, back at the hostel, the kids spent the evening preparing for their final fight. They knew what they were doing. They were experienced. We had a plan. I had no doubts about how well they would do. Noreen and I, meanwhile, spent the evening trying to figure out where to take them for a special dinner after the tournament and how the heck we were going to get home on Sunday.

Epilogue: What the Trip Was Really About

The easy way to describe this trip is to say that Team USA went to Switzerland to compete in an international physics tournament. That is true, but it is also too small.

This trip was about Val standing in rooms full of international competitors and proving that her work could stand next to anyone's. It was about Bridget taking on Germany, getting tested, and holding her own. It was about Irene being placed in an impossible situation, swallowing fear, and presenting anyway. It was

about Ivey finding her voice, not just quietly contributing in the background, but standing up, speaking clearly, making eye contact, and showing everyone the student she has become. It was about Kurt bringing his German-team experience into Team USA and then becoming, in the best possible way, fully one of us.

It was also about Noreen and me trying to feed everyone, move everyone, encourage everyone, and occasionally violate Swiss hotel policies with enough charm that nobody called the authorities.

The physics mattered. The scores mattered. The medals mattered. The students cared deeply about doing well, and so did I. But what I will remember most is watching them become a team. They picked each other up. They argued. They processed disappointment. They worked when they were tired. They laughed. They ate ice cream. They hiked mountains. They backed Irene when she needed them. They listened to Kurt. They trusted Val. They celebrated Bridget. They made room for Ivey. They became something more than five smart kids with five physics problems.

They became Team USA.

And yes, somewhere along the way, they probably made me a little better too.